

# Lecture 5: K Nearest Neighbor

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# Overview

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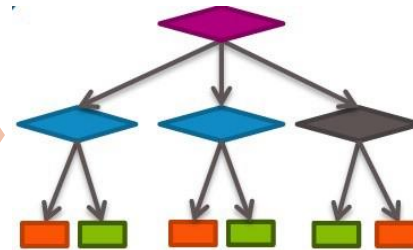
- Understanding K Nearest Neighbor
  - The K-NN algorithm
    - Measuring similarity with distance
    - Choosing an appropriate K
- Overfitting and Its Avoidance

# K Nearest Neighbor

## Information of Cars

- Auction
- Color
- MMRCurrentAuctionAveragePrice
- Size
- TopThreeAmericanName
- VehBCost
- VehicleAge
- VehOdo
- WarrantyCost
- WheelType

Predictors (X)



**Classifier**

Decision Tree, Naïve Bayes or  
k Nearest Neighbor

IsBadBuy

Target variable  
/Label (Y)

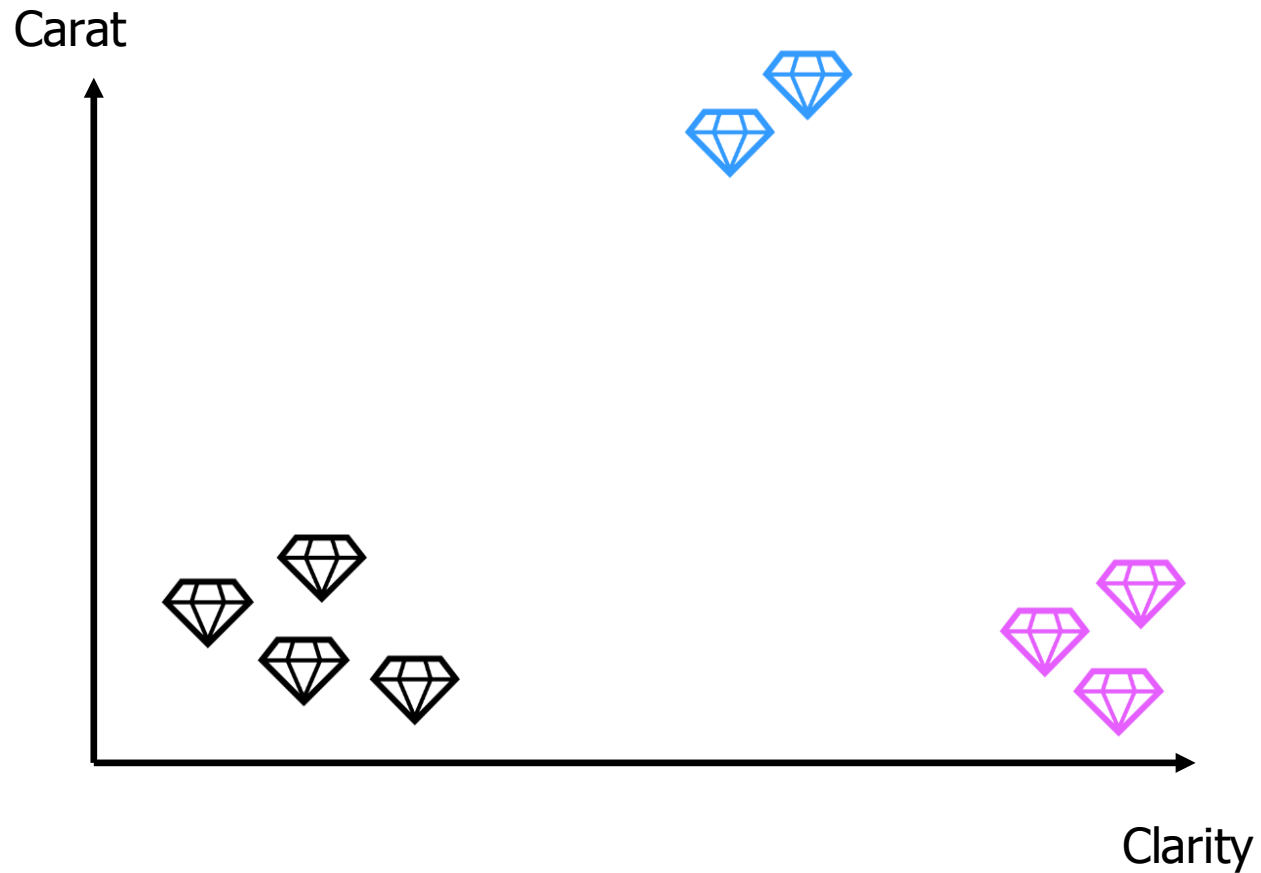
# A Simple Classification Model – Nearest Neighbor

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- Diamond professionals use the grading system developed by GIA in the 1950s, which established the use of four important factors to describe and classify diamonds: Clarity, Color, Cut, and Carat Weight.

# Nearest Neighbor



C



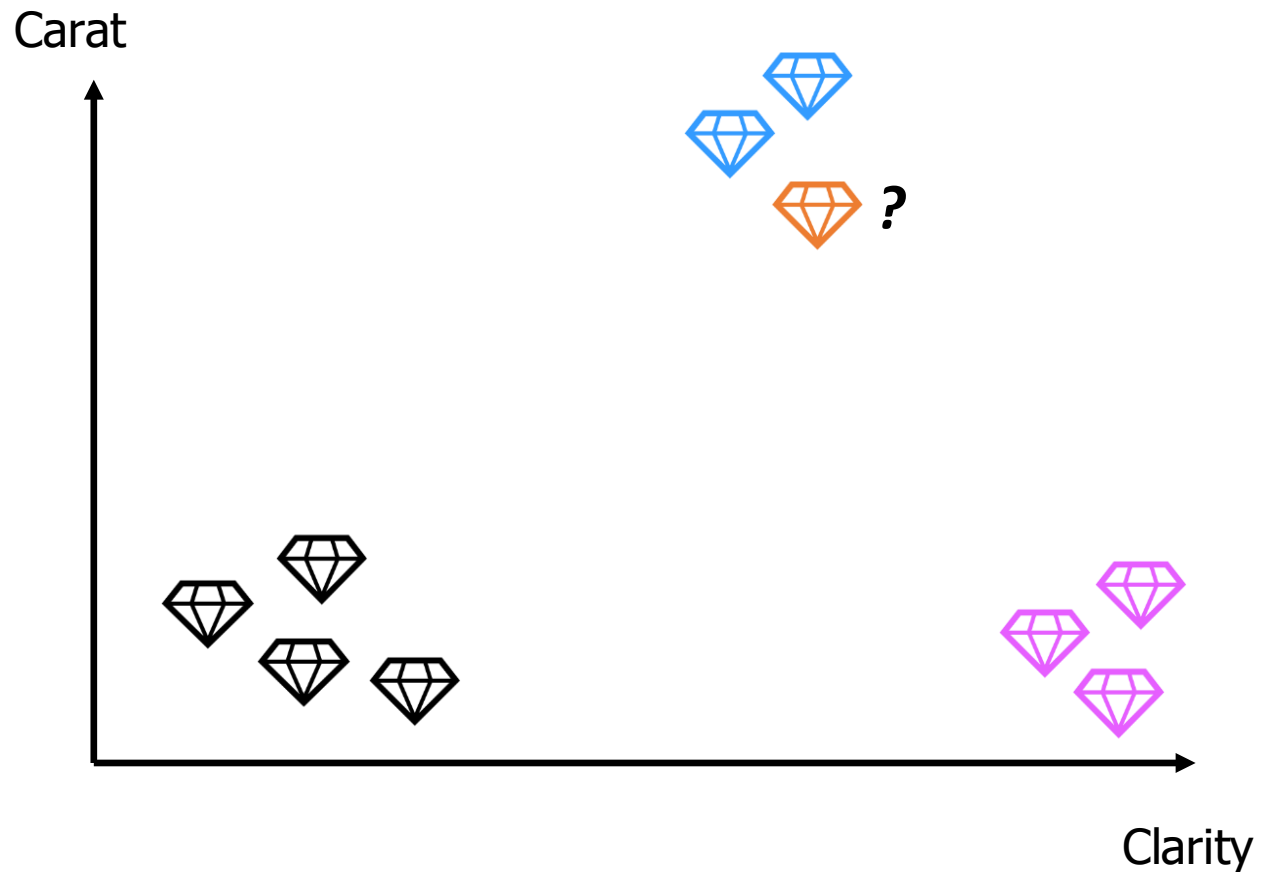
B



A

|          | Clarity | Carat | Class |
|----------|---------|-------|-------|
| Diamond1 | 8       | 3     | A     |
| Diamond2 | 7       | 2.8   | A     |
| Diamond3 | 10      | 1     | B     |
| Diamond4 | 9.5     | 0.8   | B     |
| Diamond5 | 9       | 0.5   | B     |
| Diamond6 | 3       | 0.5   | C     |
| Diamond7 | 2       | 1     | C     |
| Diamond8 | 2       | 0.5   | C     |
| Diamond9 | 1       | 0.7   | C     |

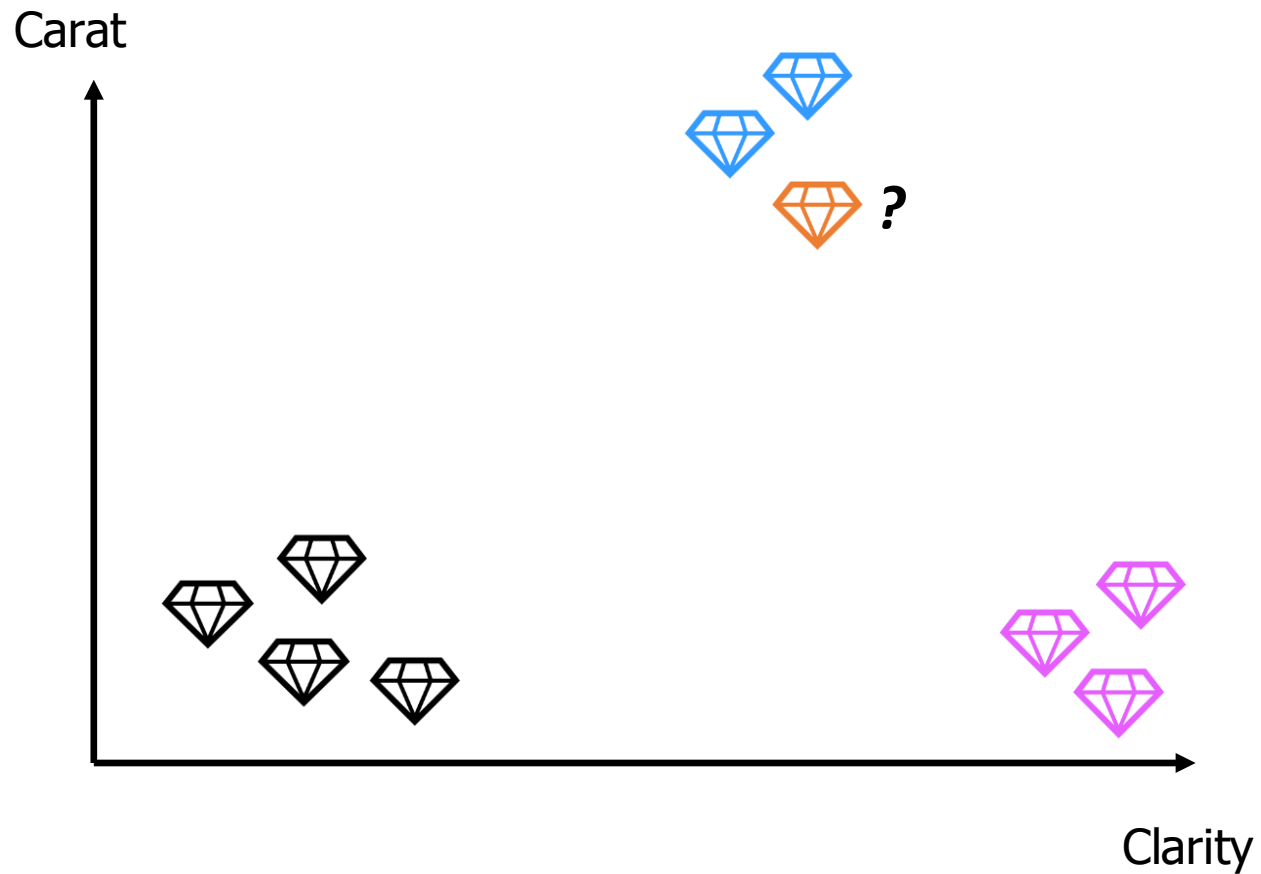
# Nearest Neighbor



- Classification Task: When a new diamond comes, how can classify it to A, B, or C?

|          | Clarity | Carat | Class |
|----------|---------|-------|-------|
| Diamond1 | 8       | 3     | A     |
| Diamond2 | 7       | 2.8   | A     |
| Diamond3 | 10      | 1     | B     |
| Diamond4 | 9.5     | 0.8   | B     |
| Diamond5 | 9       | 0.5   | B     |
| Diamond6 | 3       | 0.5   | C     |
| Diamond7 | 2       | 1     | C     |
| Diamond8 | 2       | 0.5   | C     |
| Diamond9 | 1       | 0.7   | C     |

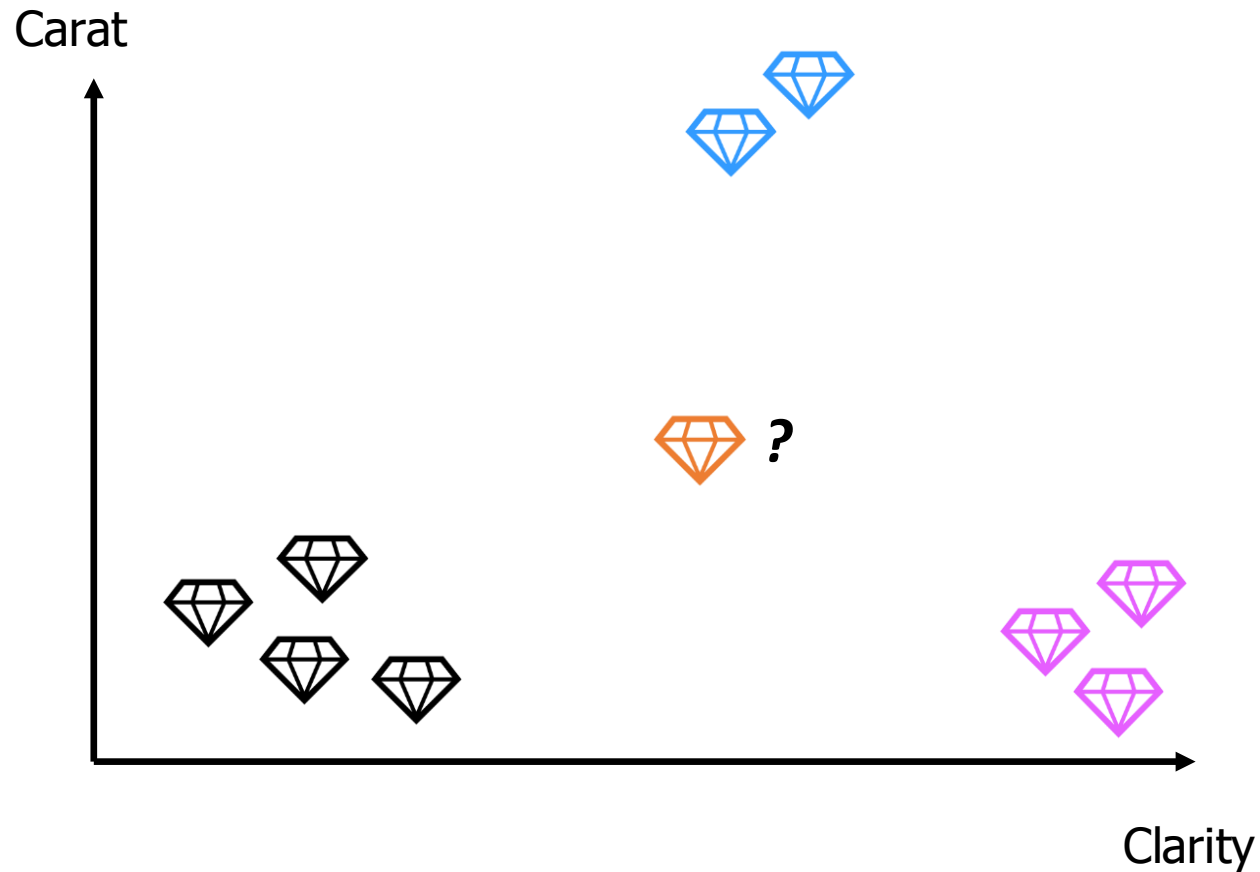
# Nearest Neighbor



is classified as A,  
because it is more  
similar (closer) to the  
diamonds in class A.

# Nearest Neighbor

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- It is hard to classify which category it belongs to.



# Measuring Similarity with Distance

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- We use **distance function**, or a formula that measures the similarity between the two instances.
  - An instance (e.g., a customer) has a list of variables
    - e.g., attributes of a customer such as age, spending, gender etc.
- To measure similarity between two instances, we measure similarity between these instances' variable values based on a distance function.
  - Input: variable values of two instances
  - Output: distance

# Measuring Similarity with Distance

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- There are many different ways to calculate distance. The most commonly used measures are

- Euclidean distance

$$d(X, Y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2}$$

where  $x_1 \dots x_n$  are values of variables of instance X

and  $y_1 \dots y_n$  are values of variables of instance Y

# Measuring Similarity with Distance

- E.g., Euclidean distance (Tom, Jack)=

$$\sqrt{(32 - 35)^2 + (5400 - 6000)^2} = 600.008$$

| NAME | AGE | SPENDING(\$) |
|------|-----|--------------|
| SUE  | 21  | 2300         |
| CARL | 27  | 2600         |
| TOM  | 32  | 5400         |
| JACK | 35  | 6000         |
| DAN  | 44  | 6200         |
| JILL | 50  | 7000         |



The distance is dominated  
by the second variable

# Measuring Similarity with Distance

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- Variables are typically transformed to a **standard range** prior to applying the k-NN algorithm.
  - The rationale for this step is that the distance formula is highly dependent on how variables are measured.
  - If certain variables have a much larger range of values than the others, the distance measurements will be strongly dominated by the variables with larger ranges.

- **min-max normalization**

$$X_{new} = \frac{X - \min(X)}{\max(X) - \min(X)}$$

With min-max normalization, each feature contributes equally to the distance formula

# Measuring Similarity with Distance

| NAME | AGE | SPENDING(\$) |
|------|-----|--------------|
|------|-----|--------------|

|     |    |      |
|-----|----|------|
| SUE | 21 | 2300 |
|-----|----|------|

|      |    |      |
|------|----|------|
| CARL | 27 | 2600 |
|------|----|------|

|     |    |      |
|-----|----|------|
| TOM | 32 | 5400 |
|-----|----|------|

|      |    |      |
|------|----|------|
| JACK | 35 | 6000 |
|------|----|------|

|     |    |      |
|-----|----|------|
| DAN | 44 | 6200 |
|-----|----|------|

|      |    |      |
|------|----|------|
| JILL | 50 | 7000 |
|------|----|------|



| NAME | AGE | SPENDING(\$) |
|------|-----|--------------|
|------|-----|--------------|

|     |   |   |
|-----|---|---|
| SUE | 0 | 0 |
|-----|---|---|

|      |       |       |
|------|-------|-------|
| CARL | 0.207 | 0.064 |
|------|-------|-------|

|     |       |       |
|-----|-------|-------|
| TOM | 0.379 | 0.660 |
|-----|-------|-------|

|      |       |       |
|------|-------|-------|
| JACK | 0.483 | 0.787 |
|------|-------|-------|

|     |       |       |
|-----|-------|-------|
| DAN | 0.793 | 0.830 |
|-----|-------|-------|

|      |   |   |
|------|---|---|
| JILL | 1 | 1 |
|------|---|---|

- Euclidean distance (Tom, Jack)=

$$\sqrt{(0.379 - 0.483)^2 + (0.660 - 0.787)^2} = 0.164$$

# Measuring Similarity with Distance

- The Euclidean distance is not defined for categorical variables.
- Convert categorical variables into numeric format – **dummy coding**
  - **Create dummy variables to replace the original categorical variable.**
  - Dummy coding for a binary variable: gender

$$\text{male} = \begin{cases} 1 & \text{if } x = \text{male} \\ 0 & \text{otherwise} \end{cases}$$

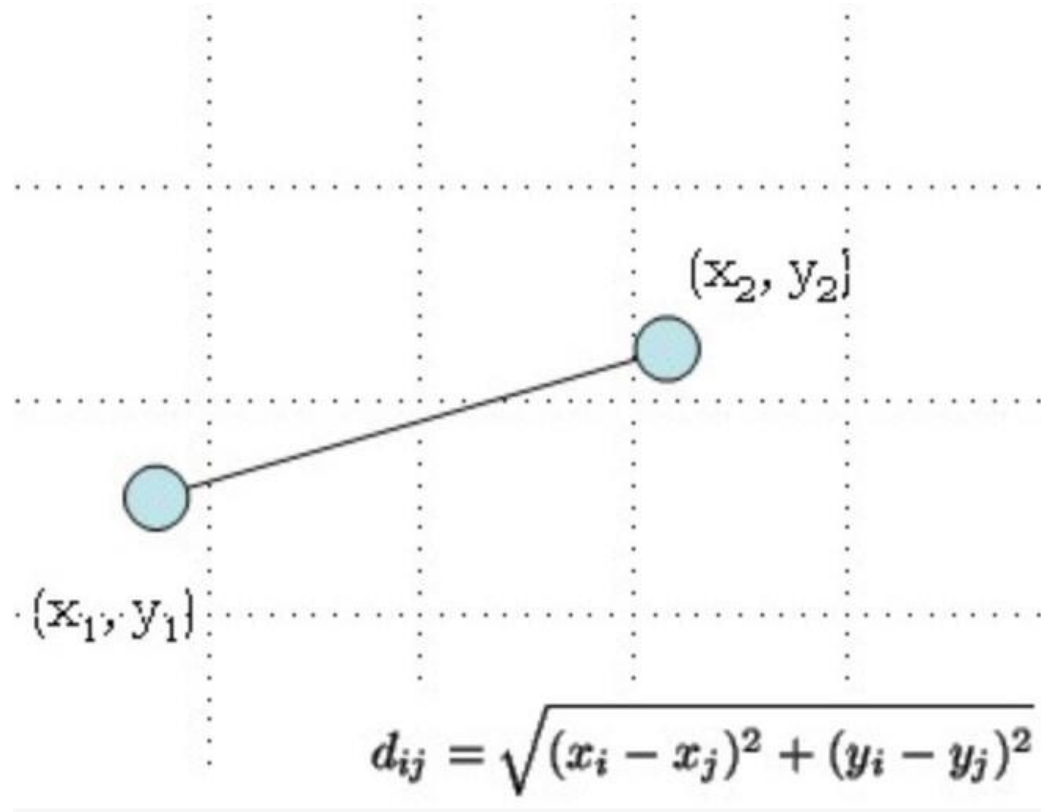
| Color: Red, Pink, Blue |       | <i>n</i> categories, <i>n</i> -1 dummy variables |           |            |
|------------------------|-------|--------------------------------------------------|-----------|------------|
|                        | Color |                                                  | Color_Red | Color_Pink |
| Diamond1               | Red   | Diamond1                                         | 1         | 0          |
| Diamond2               | Pink  | Diamond2                                         | 0         | 1          |
| Diamond3               | Blue  | Diamond3                                         | 0         | 0          |

# Measuring Similarity with Distance

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| NAME | AGE | SPENDING(\$) | GENDER |                                                                                     | NAME | AGE   | SPENDING(\$) | GENDER |
|------|-----|--------------|--------|-------------------------------------------------------------------------------------|------|-------|--------------|--------|
| SUE  | 21  | 2300         | F      |  | SUE  | 0     | 0            | 1      |
| CARL | 27  | 2600         | M      |                                                                                     | CARL | 0.207 | 0.064        | 0      |
| TOM  | 32  | 5400         | M      |                                                                                     | TOM  | 0.379 | 0.660        | 0      |
| JACK | 35  | 6000         | M      |                                                                                     | JACK | 0.483 | 0.787        | 0      |
| DAN  | 44  | 6200         | M      |                                                                                     | DAN  | 0.793 | 0.830        | 0      |
| JILL | 50  | 7000         | F      |                                                                                     | JILL | 1     | 1            | 1      |

# Nearest Neighbor



(clarity:7, carat:2.8) → (clarity: $\frac{7-1}{10-1}$ ,  
carat: $\frac{2.8-0.5}{3-0.5}$ )

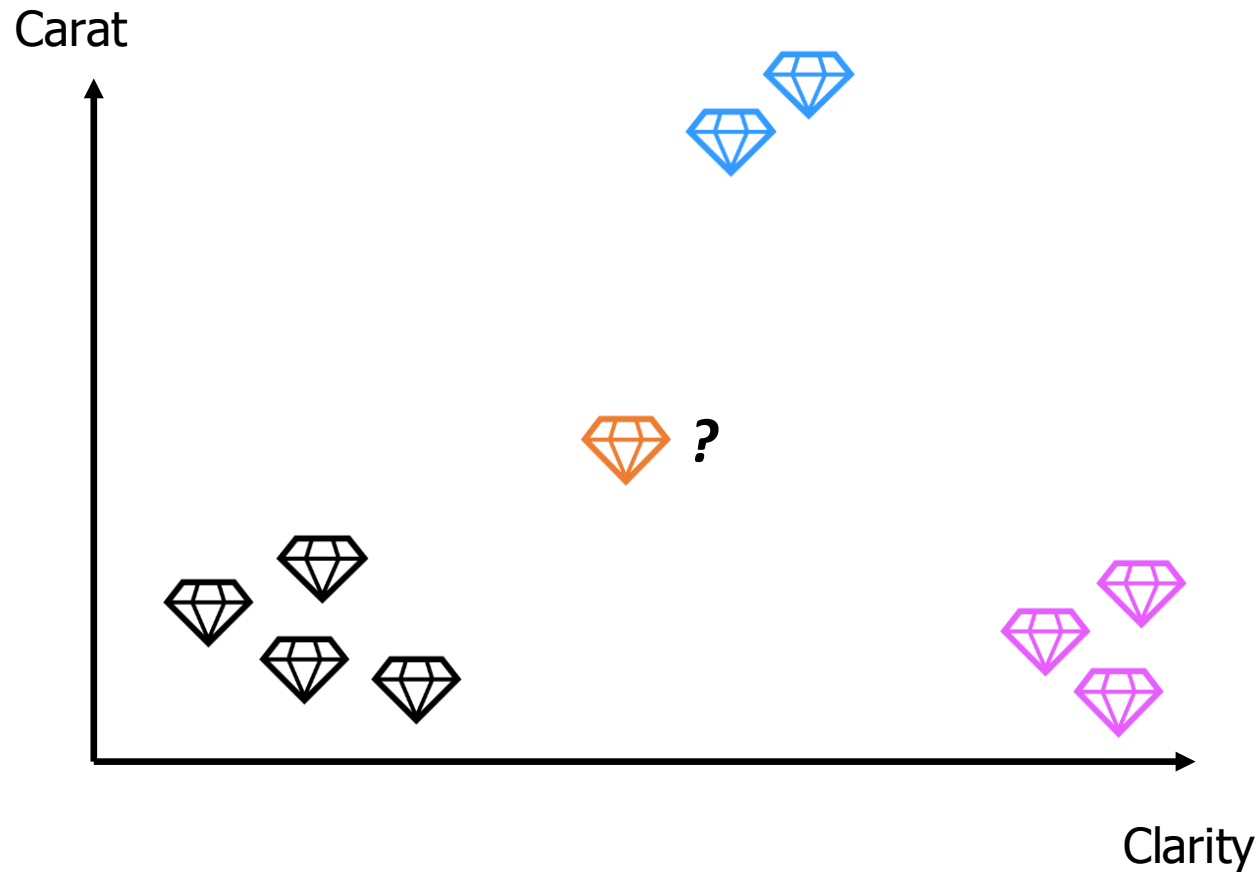


(clarity:8, carat:2.5) → (clarity: $\frac{8-1}{10-1}$ ,  
carat: $\frac{2.5-0.5}{3-0.5}$ )

$$\text{Distance} = \sqrt{(0.667 - 0.778)^2 + (0.92 - 0.8)^2} = \sqrt{0.012 + 0.0144} \approx 0.162$$

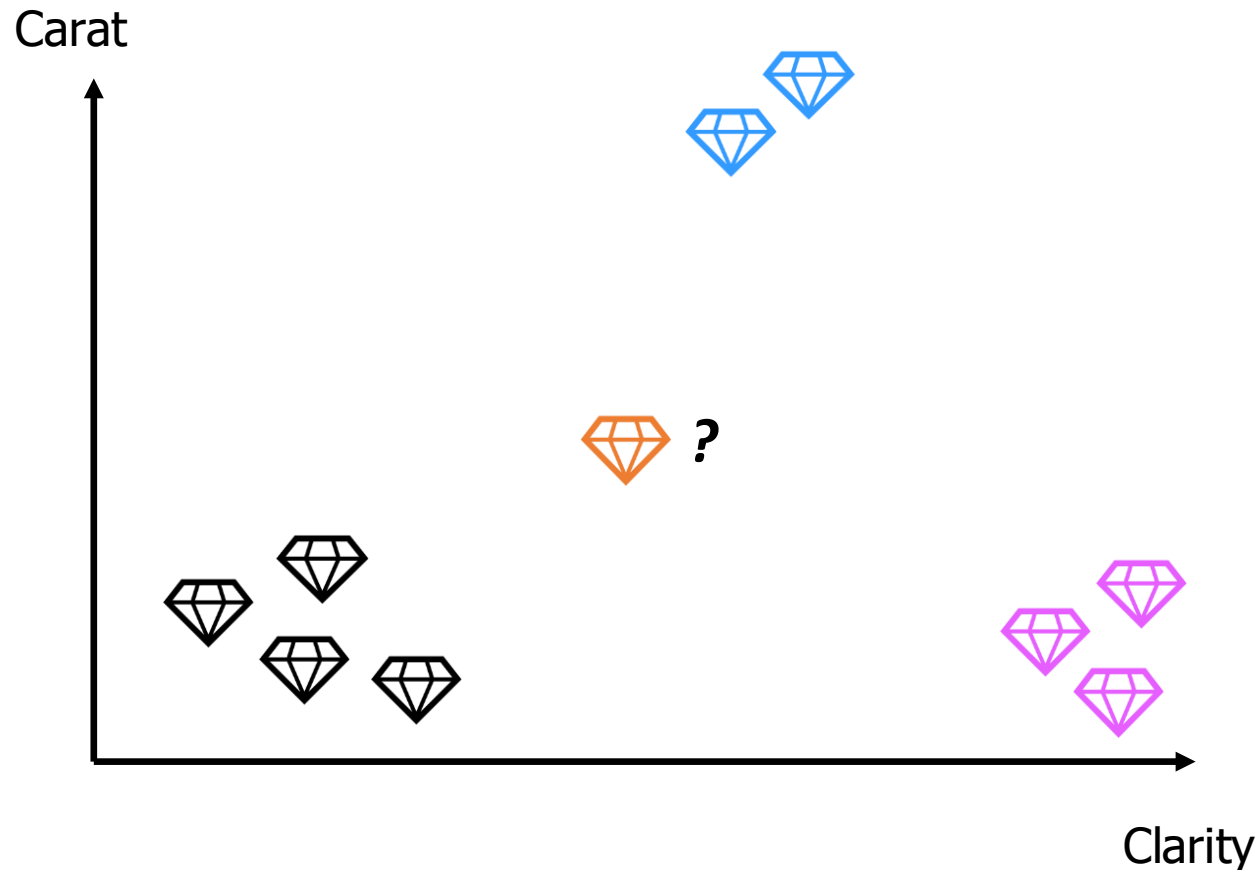


# Nearest Neighbor



|           | Clarity | Carat | Class |
|-----------|---------|-------|-------|
| Diamond1  | 8       | 3     | A     |
| Diamond2  | 7       | 2.8   | A     |
| Diamond3  | 10      | 1     | B     |
| Diamond4  | 9.5     | 0.8   | B     |
| Diamond5  | 9       | 0.5   | B     |
| Diamond6  | 3       | 0.5   | C     |
| Diamond7  | 2       | 1     | C     |
| Diamond8  | 2       | 0.5   | C     |
| Diamond9  | 1       | 0.7   | C     |
| Diamond10 | 4       | 1.5   | ???   |

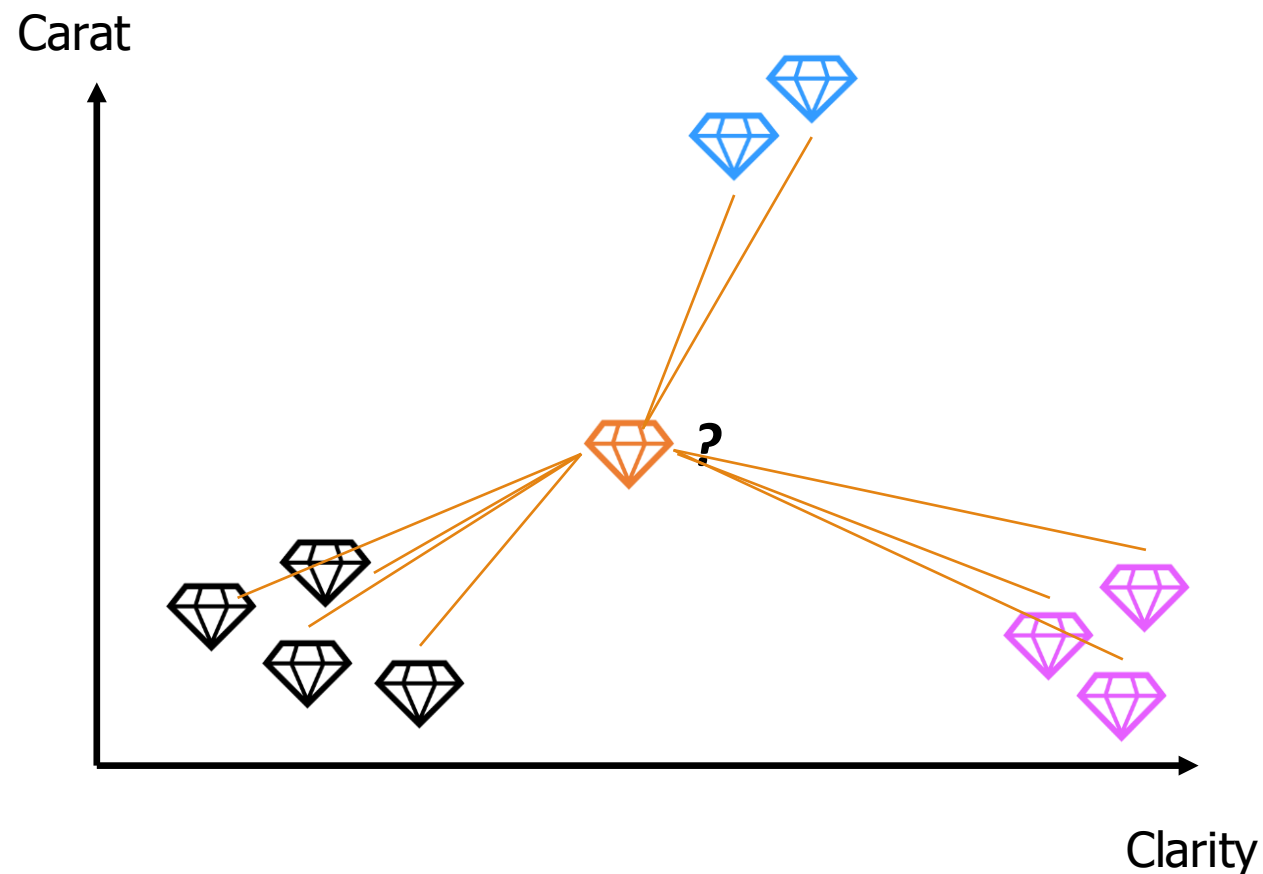
# Nearest Neighbor



## Step1. min-max Normalization

|           | Clarity | Carat | Class |
|-----------|---------|-------|-------|
| Diamond1  | 0.778   | 1     | A     |
| Diamond2  | 0.667   | 0.92  | A     |
| Diamond3  | 1       | 0.2   | B     |
| Diamond4  | 0.944   | 0.12  | B     |
| Diamond5  | 0.889   | 0     | B     |
| Diamond6  | 0.222   | 0     | C     |
| Diamond7  | 0.111   | 0.2   | C     |
| Diamond8  | 0.111   | 0     | C     |
| Diamond9  | 0       | 0.08  | C     |
| Diamond10 | 0.333   | 0.4   | ???   |

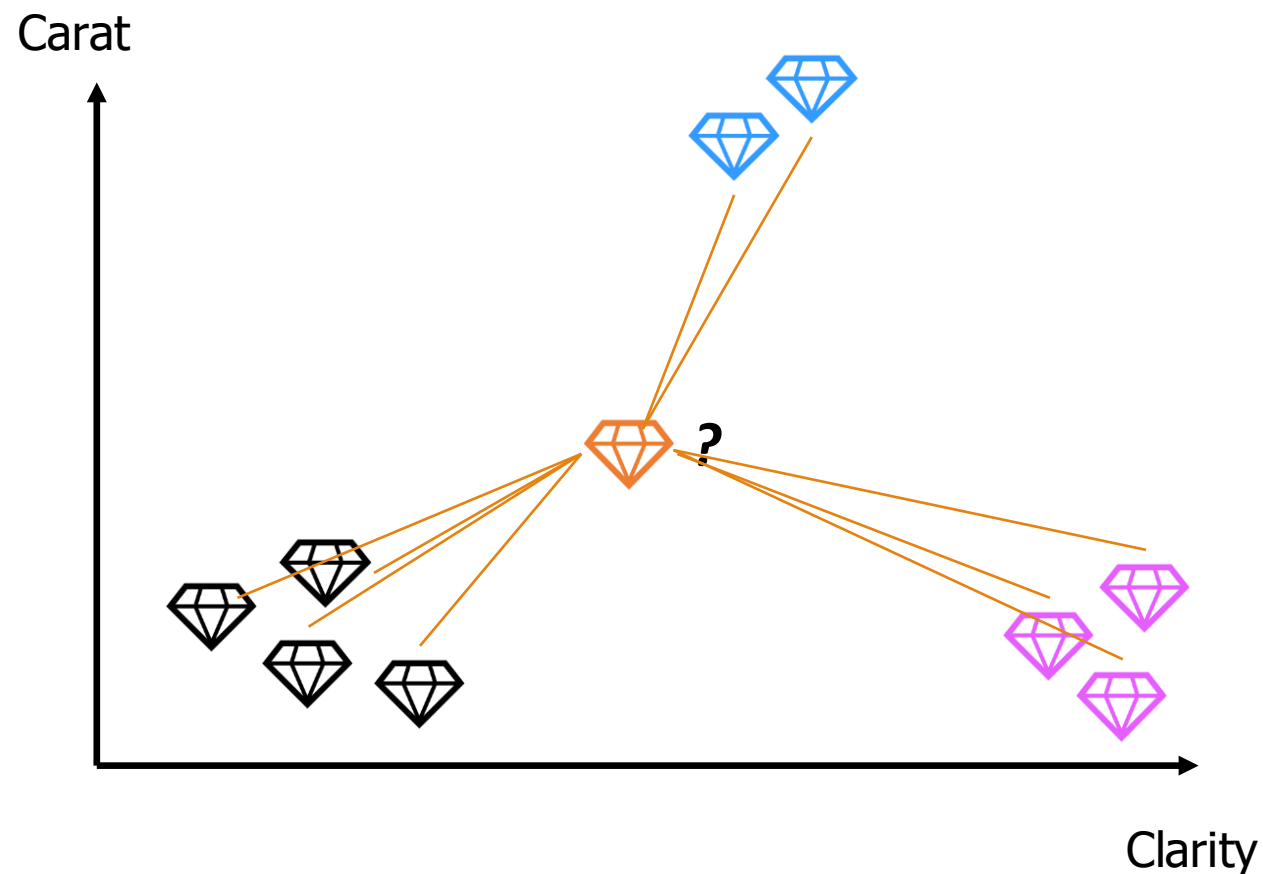
# Nearest Neighbor



## Step2. Calculate distance

|           | Clarity | Carat | Class | Distance |
|-----------|---------|-------|-------|----------|
| Diamond1  | 0.778   | 1     | A     | 0.747    |
| Diamond2  | 0.667   | 0.92  | A     | 0.618    |
| Diamond3  | 1       | 0.2   | B     | 0.696    |
| Diamond4  | 0.944   | 0.12  | B     | 0.672    |
| Diamond5  | 0.889   | 0     | B     | 0.685    |
| Diamond6  | 0.222   | 0     | C     | 0.415    |
| Diamond7  | 0.111   | 0.2   | C     | 0.299    |
| Diamond8  | 0.111   | 0     | C     | 0.458    |
| Diamond9  | 0       | 0.08  | C     | 0.462    |
| Diamond10 | 0.333   | 0.4   | ???   | 0        |

# Nearest Neighbor



**Step3. Identify the  $k$  nearest neighbors; make prediction based on the class of  $k$  nearest neighbors**

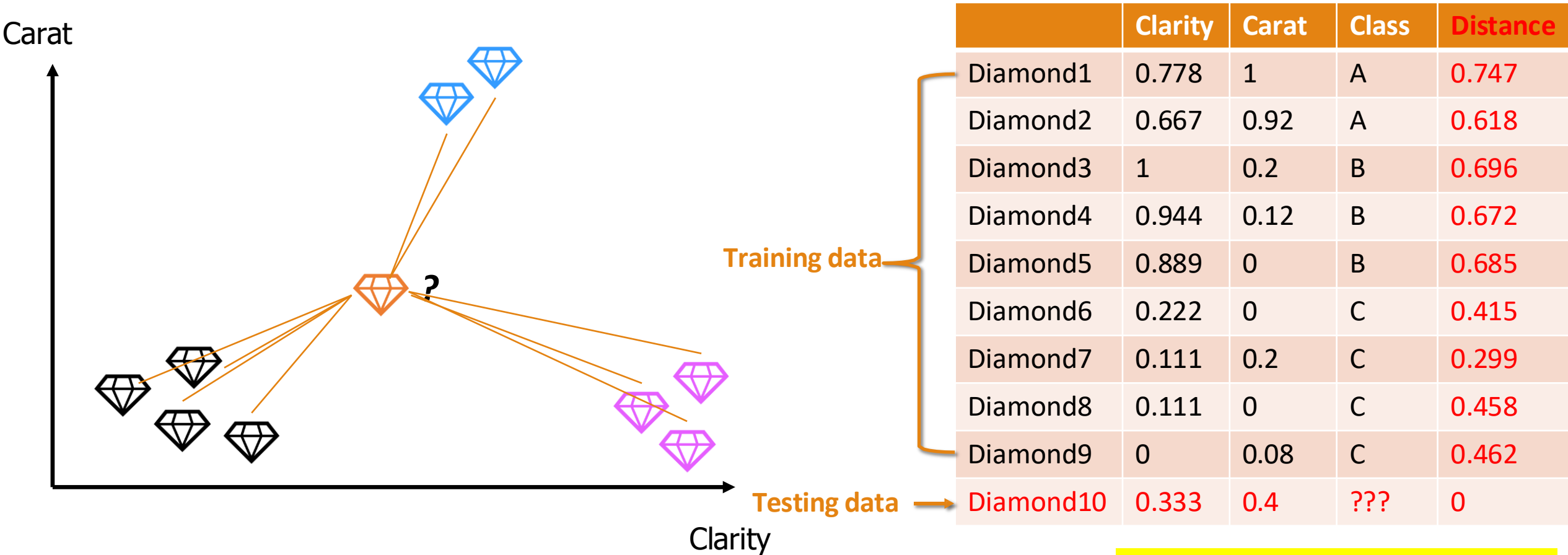
|           | Clarity | Carat | Class | Distance |
|-----------|---------|-------|-------|----------|
| Diamond1  | 0.778   | 1     | A     | 0.747    |
| Diamond2  | 0.667   | 0.92  | A     | 0.618    |
| Diamond3  | 1       | 0.2   | B     | 0.696    |
| Diamond4  | 0.944   | 0.12  | B     | 0.672    |
| Diamond5  | 0.889   | 0     | B     | 0.685    |
| Diamond6  | 0.222   | 0     | C     | 0.415    |
| Diamond7  | 0.111   | 0.2   | C     | 0.299    |
| Diamond8  | 0.111   | 0     | C     | 0.458    |
| Diamond9  | 0       | 0.08  | C     | 0.462    |
| Diamond10 | 0.333   | 0.4   | ???   | 0        |

# K Nearest Neighbor Algorithm

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- The k-NN algorithm uses information about an example's  $k$ -nearest neighbors to classify unlabeled examples.
  - The letter  $k$  is a variable term implying that any number of nearest neighbors could be used.
  - A training dataset made up of examples that have been classified into several categories.
  - For each unlabeled record in the test dataset, k-NN identifies  $k$  records in the training data that are the "nearest" in similarity.
  - The unlabeled test instance is assigned the class of the majority of the  $k$  nearest neighbors.

# Nearest Neighbor



What is the largest/smallest  $k$  value?

# Choosing an Appropriate $k$

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- The decision of how many neighbors to use for  $k$ -NN determines how well the model will generalize to future data.
- Choosing a large  $k$  reduces the impact or variance caused by noisy data, but can bias the learner so that it runs the risk of ignoring small, but important patterns. **UNDERFITTING**  $K = \text{total number of instances in the training data}$
- On the opposite extreme, using a single nearest neighbor allows the noisy data or outliers to unduly influence the classification of examples. **OVERFITTING**  $K = 1$

# K Nearest Neighbor

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- Training phase:
  - Import and store training data in the “memory”
  - No actual “model” is built
  - Fast but memory intensive
- Prediction phase:
  - The main work and time consuming
- Common characterization
  - Instance-based learning algorithms
  - Memory-based learning algorithms
  - Lazy learners



# K Nearest Neighbor

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- Issues with K Nearest Neighbor
  - There is a problem with having too many attributes, or many that are irrelevant to the similarity judgment.
  - Computational efficiency
  - Does not produce a model, limiting the ability to understand how the features are related to the class
  - Require selection of an appropriate  $k$

# KNN Distance Metrics in scikit-learn (Numeric)

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## Minkowski (L<sub>p</sub>)

- Formula:  $d(x,y) = (\sum |x_i - y_i|^p)^{1/p}$  Code: `metric='minkowski', p=3`
- general-purpose; tune p (p=1,2 most common)

## Euclidean (L<sub>2</sub>)

- Formula:  $d(x,y) = \sqrt{\sum (x_i - y_i)^2}$  Code: `metric='euclidean'` (or `metric='minkowski', p=2`)
- standard for numeric features after scaling

## Manhattan (L<sub>1</sub>)

- Formula:  $d(x,y) = \sum |x_i - y_i|$  Code: `metric='manhattan'` (or `metric='minkowski', p=1`)
- more robust than L<sub>2</sub>; often better in higher dimensions

## Chebyshev (L<sub>∞</sub>)

- Formula:  $d(x,y) = \max |x_i - y_i|$  Code: `metric='chebyshev'`
- when the largest single-feature difference should dominate

## Standardized Euclidean (SEuclidean)

- Formula:  $d(x,y) = \sqrt{\sum (x_i - y_i)^2 / v_i}$  Code: `metric='seuclidean', metric_params={'V': variances}`
- numeric features with very different variances (still scale carefully)

# KNN Distance Metrics in scikit-learn (other)

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**Cosine distance:**  $d(x,y) = 1 - (x \cdot y) / (\|x\| \|y\|)$  Code: `metric='cosine'`

- text vectors / embeddings (direction matters more than magnitude)

**Hamming distance:**  $d(x,y) = (1/n) \sum [x_i \neq y_i]$  Code: `metric='hamming'`

- binary or categorical (0/1) features; mismatch rate

**Jaccard distance (binary):**  $d(x,y) = 1 - |x \cap y| / |x \cup y|$  Code: `metric='jaccard'`

- sparse binary “set-like” features; presence/absence overlap

**Mahalanobis distance:**  $d(x,y) = \sqrt{(x-y)^T S^{-1} (x-y)}$

- Code: `metric='mahalanobis', metric_params={'V': cov_matrix}`
- correlated numeric features; accounts for correlation

**Canberra distance:**  $d(x,y) = \sum |x_i - y_i| / (|x_i| + |y_i|)$  Code: `metric='canberra'`

- sparse/count-like data; emphasizes relative differences

**Bray-Curtis distance:**  $d(x,y) = \sum |x_i - y_i| / \sum |x_i + y_i|$  Code: `metric='braycurtis'`

- proportion/composition-like comparisons

# K Nearest Neighbor

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- KNN can be easily adjusted to do regression.  
Think how?